Experiment 2: DDL Commands with Constraints

AIM

To create tables using PRIMARY KEY, FOREIGN KEY, UNIQUE, CHECK, DEFAULT, and NOT NULL constraints in MySQL.

Outputs:

```
mysql> CREATE TABLE Department ( dept_id INT PRIMARY KEY, dept_name VARCHAR(30) UNIQUE NOT NULL);
Query OK, 0 rows affected (0.08 sec)

mysql> INSERT INTO Department
    -> VALUES (10, 'CSE'),(20, 'IT');
Query OK, 2 rows affected (0.02 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Department;
+---------+-----------+
| dept_id | dept_name |
+---------+-----------+
|      10 | CSE       |
|      20 | IT        |
+---------+-----------+
2 rows in set (0.01 sec)
```

```
mysql> CREATE TABLE Student (student_id INT PRIMARY KEY,name VARCHAR(50) NOT NULL,email VARCHAR(100) UNIQUE,age INT CHECK (age >= 18), city VARC
HAR(30) DEFAULT 'Chennai',dept_id INT,FOREIGN KEY (dept_id)REFERENCES Department(dept_id));
Query OK, 0 rows affected (0.10 sec)

mysql> INSERT INTO Student(student_id, name, email, age, dept_id)
    -> VALUES (1, 'Bhavya', 'bhavya@gmail.com', 20, 10),(2, 'Ravi', 'ravi@gmail.com', 21, 20);
Query OK, 2 rows affected (0.01 sec)
Records: 2  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Student;
+------------+--------+-----------------+------+---------+---------+
| student_id | name   | email           | age  | city    | dept_id |
+------------+--------+-----------------+------+---------+---------+
|          1 | Bhavya | bhavya@gmail.com |   20 | Chennai |      10 |
|          2 | Ravi   | ravi@gmail.com   |   21 | Chennai |      20 |
+------------+--------+-----------------+------+---------+---------+
2 rows in set (0.00 sec)
```

```
mysql> INSERT INTO Student
    -> VALUES (1, 'Kiran', 'kiran@gmail.com', 22, 'Chennai', 10);
ERROR 1062 (23000): Duplicate entry '1' for key 'student.PRIMARY'
```

```
mysql>  INSERT INTO Student(student_id, name, email, age, dept_id)
    -> VALUES (3, NULL, 'test@gmail.com', 20, 10);
ERROR 1048 (23000): Column 'name' cannot be null
```

```
mysql> INSERT INTO Student(student_id, name, email, age, dept_id)
    -> VALUES (3, 'Kiran', 'bhavya@gmail.com', 22, 10);
ERROR 1062 (23000): Duplicate entry 'bhavya@gmail.com' for key 'student.email'
mysql>  INSERT INTO Student(student_id, name, email, age, dept_id)
    -> VALUES (3, 'Kiran', 'kiran@gmail.com', 16, 10);
ERROR 3819 (HY000): Check constraint 'student_chk_1' is violated.
```

```
mysql>  INSERT INTO Student(student_id, name, email, age, dept_id)
    -> VALUES (3, 'Kiran', 'kiran@gmail.com', 22, 99);
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails (`school_db`.`student`, CONSTRAINT `student_ibfk_1` FOREIGN
 KEY (`dept_id`) REFERENCES `department` (`dept_id`))
mysql>
```

RESULT

Thus, the tables were created with PRIMARY KEY, FOREIGN KEY, UNIQUE, CHECK, DEFAULT, and NOT NULL constraints. Valid records were inserted and displayed, and invalid records can be used to verify that the constraints reject violations. The city column takes the default value 'Chennai' when omitted.